

Task_2

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait()`. 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using `fork()` system call.
- b) Read the user input as shell command.
- c) Use a system call `exec()` to execute the shell command.
- d) Use a `wait()` system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

```
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ nano prctical_02.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ gcc prctical_02.c -o prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ ./prctical_02
Child Process Executing
Parent Process After Waiting
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ cat prctical_02.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid == 0)
    {
        printf("Child Process Executing\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Process After Waiting\n");
    }

    return 0;
}
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ |

saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ nano prctical_02.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ gcc prctical_02.c -o prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ ./prctical_02
Parent PID : 15969
Child PID : 15970
Child PID : 15970
Parent PID : 15969
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ cat prctical_02.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }

    return 0;
}
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ |
```

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```
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ nano prctical_02.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ gcc prctical_02.c -o prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ ./prctical_02
Enter a Linux command: pwd
/home/saireddy/OSSP/practical/prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ cat prctical_02.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    execlp(command, command, NULL);

    printf("Command execution failed.\n");

    return 0;
}
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ |

saireddy@DESKTOP-HKQRL9G:~$ cd OSSP
saireddy@DESKTOP-HKQRL9G:~/OSSP$ cd practical
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ cd prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ nanao prctical_02.c
Command 'nanao' not found, did you mean:
  command 'nano' from deb nano (7.2-2ubuntu0.2)
Try: sudo apt install <deb name>
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ nano prctical_02.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ gcc prctical_02.c -o prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ ./practical_02
-bash: ./practical_02: No such file or directory
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ ./prctical_02
Enter a Linux command: ls
Command entered: ls
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ cat prctical_02.c
#include <stdio.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    printf("Command entered: %s\n", command);

    return 0;
}
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ |
```

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```
saireddy@DESKTOP-HKQRL9G:~$ cd OSSP
saireddy@DESKTOP-HKQRL9G:~/OSSP$ cd practical
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ mkdir prctical_02
Command 'mkdir' not found, did you mean:
  command 'mkdir' from deb coreutils (9.4-3ubuntu6.2)
  command 'mmkdir' from deb mblaze (1.1-1)
Try: sudo apt install <deb name>
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ mkdir prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ cd prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ nano prctical_02.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ gcc prctical_02.c -o prctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ ./prctical_01
-bash: ./prctical_01: No such file or directory
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ ./prctical_02
Parent Process Created
Child Process Created
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical/prctical_02$ cat prctical_02
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid == 0) {
        printf("Child Process Created\n");
    } else if (pid > 0) {
        printf("Parent Process Created\n");
    } else {
        printf("Fork Failed\n");
    }

    return 0;
}
```